

Rings, fields & vector spaces

linear comb's:

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$$

Exercise If you do same
 $\mathbb{Q}_6 = \{0, 1, 2, 3, 4, 5\}$
not a field.

This ring
 $2 \times 3 =$
Another important

$$\mathbb{Q}(\sqrt{2}) : a + b\sqrt{2}$$

$$(2 + 5\sqrt{2})(\dots)$$